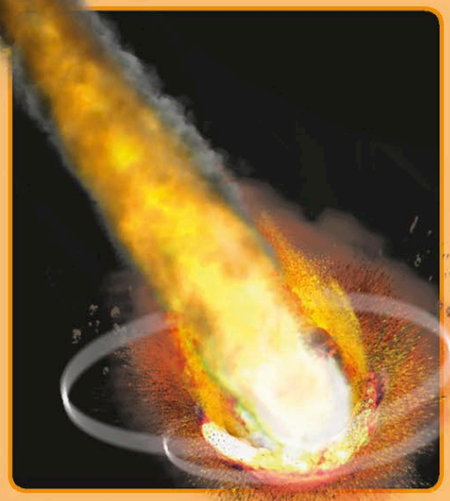


REFERENCE

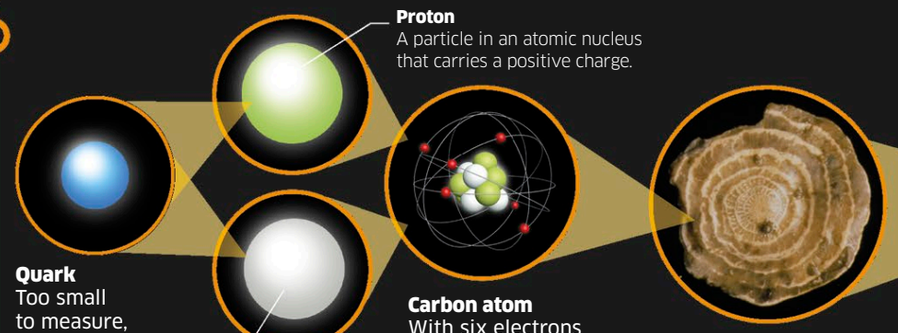
The scope of science stretches far and wide. Scientists study the vast expanse of the universe and everything within it—including the diversity of life and how it evolved. Careful observation, measurements, and experiments help scientists understand the world.

There are more **stars in the universe** than there are **grains of sand on all the beaches on Earth.**

Scale of the universe

The difference in size between the smallest and biggest things in the universe is unimaginably vast—from subatomic particles to galaxies.

No one knows how big the universe is, but it has been expanding since it formed in the Big Bang 13.7 billion years ago. The distances are so great that cosmologists measure them in terms of light-years—the distance light moves in space in a year, which is equal to 6 trillion miles (9.5 trillion kilometers)—and parts of the universe are billions of light-years apart.



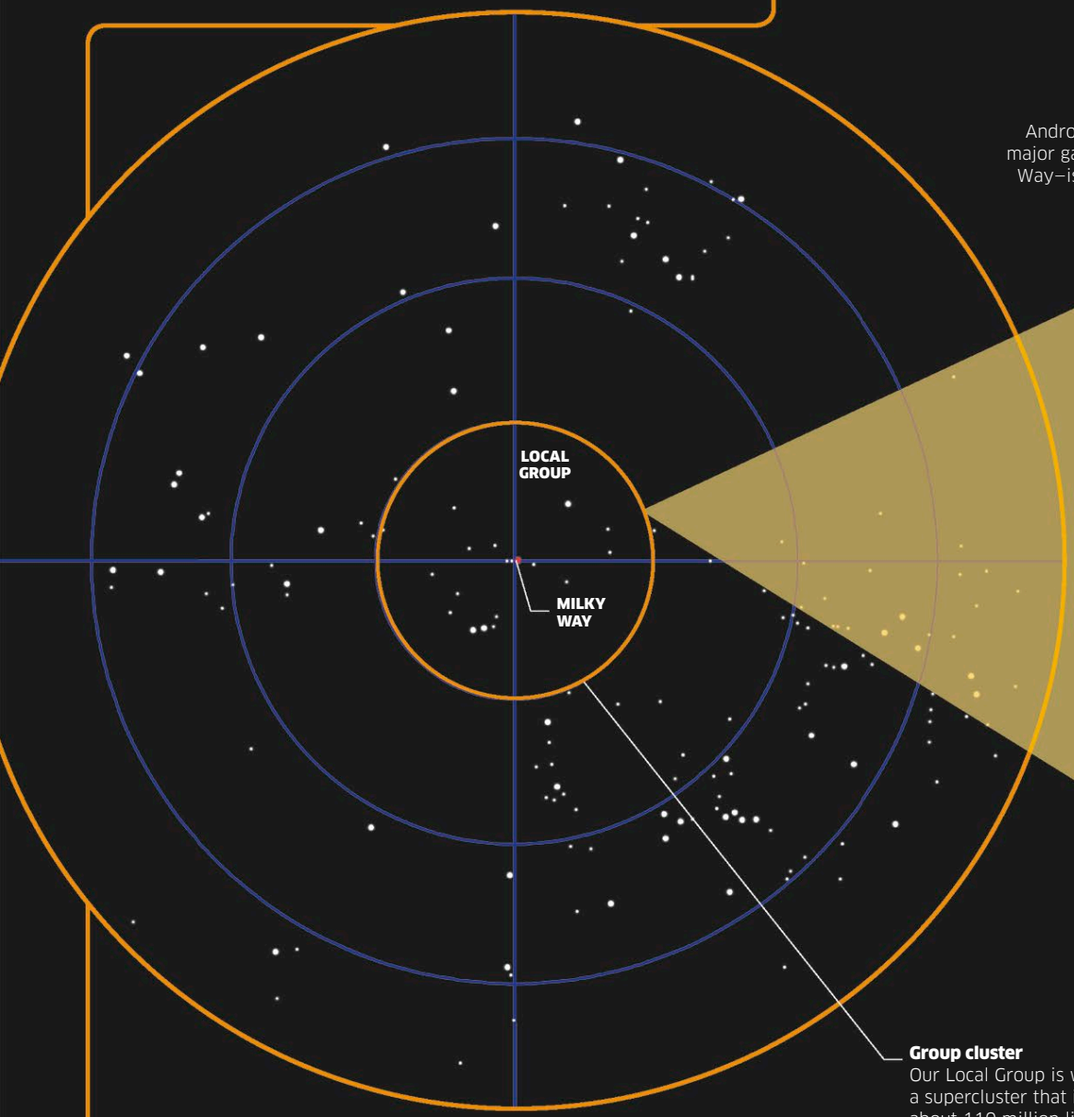
Proton
A particle in an atomic nucleus that carries a positive charge.

Quark
Too small to measure, different kinds of quarks are the subatomic particles that make up protons and neutrons.

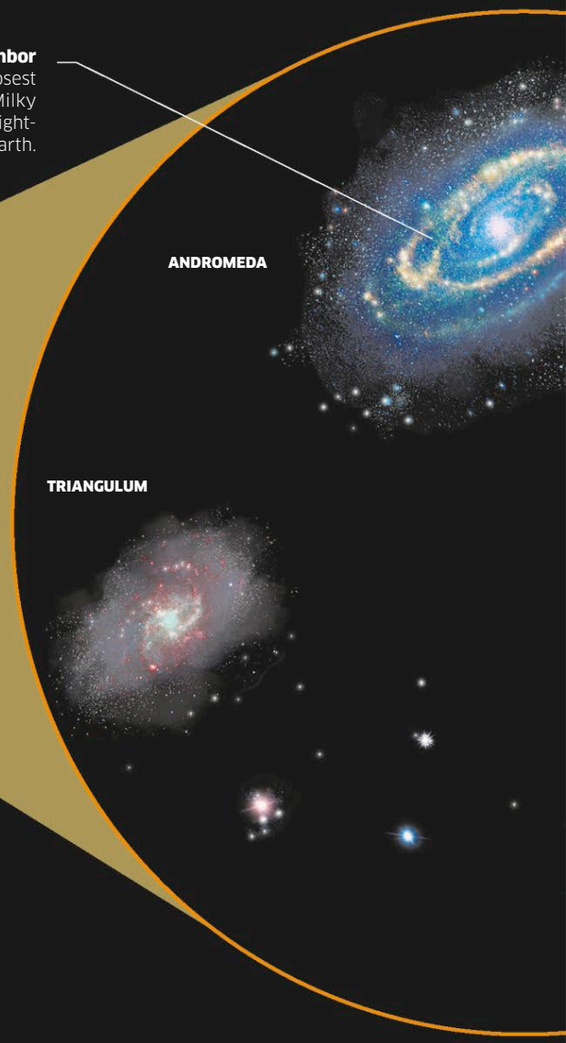
Neutron
A particle in an atomic nucleus that carries no charge.

Carbon atom
With six electrons orbiting a nucleus of six protons and six neutrons, a carbon atom is less than a billionth of a meter across.

Limestone rock
Solid mixtures of billions of tiny fossilized shells and mineral fragments make up limestone rock, containing calcium carbonate—a compound that has atoms of calcium, carbon, and oxygen.



Nearest neighbor
Andromeda—the closest major galaxy to our Milky Way—is 2.5 million light-years from Earth.



Group cluster
Our Local Group is within a supercluster that is about 110 million light-years in diameter.

Supercluster
Clusters of galaxies span a region of space ten times bigger than the Local Group. Such a supercluster can contain tens of thousands of galaxies. Our Milky Way is within the Virgo Supercluster. Scientists think there are about 10 million superclusters in the observable universe.

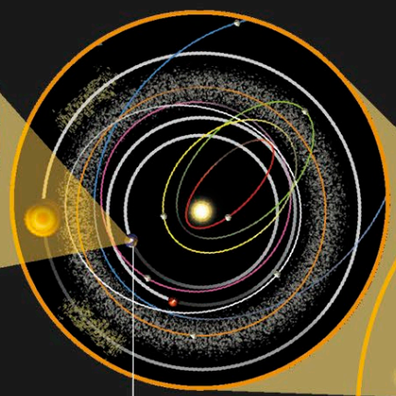
Local Group
The Milky Way is part of a so-called Local Group of about 50 galaxies that stretch across 10 million light-years of space—that's 100 times the diameter of our Milky Way. Galaxies are millions of times farther apart than the stars that are in each one. Andromeda is the biggest galaxy in our Local Group—most others are much smaller.

Mountain ranges
Moving continents push layers of rock together to create mountains.

Mount Everest
The highest land peak on planet Earth, 29,029 ft (8,848 m) high and capped with limestone, was formed over tens of millions of years.

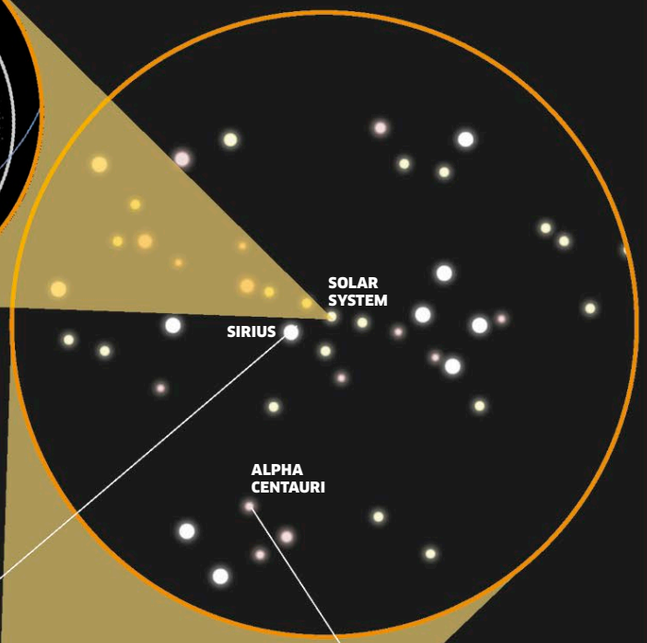
Earth
The third planet from the sun formed more than 4 billion years ago. Its diameter at the equator is 7,926 miles (12,756 km)—nearly 1,500 times Mount Everest's height.

Solar system
Eight planets orbit our sun, which has a diameter 100 times bigger than Earth's. The edge of the solar system is 122 times as far from the sun as the sun is from Earth.



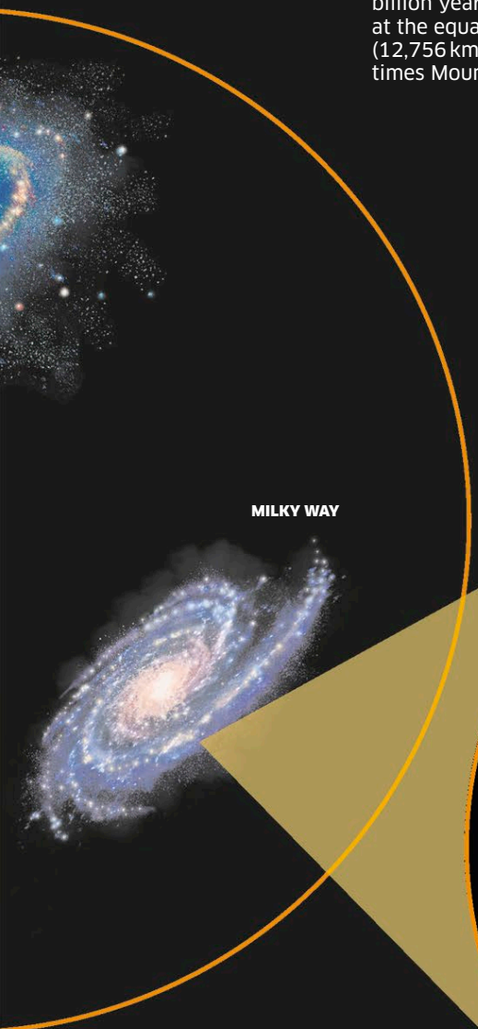
Earth to sun
The distance between Earth and the sun is 93 million miles (150 million km). This is known as one astronomical unit (AU).

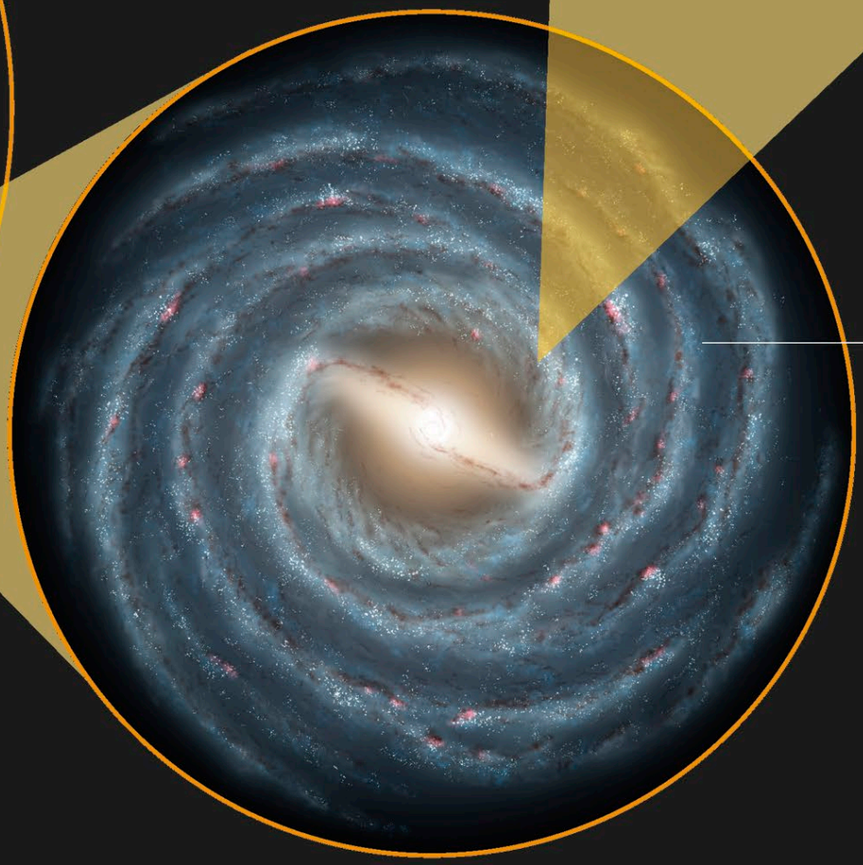
Stellar neighborhood
There are 79 star systems in our stellar neighborhood—all within a range of 20 light-years. The closest, Alpha Centauri, is 4.35 light-years away. Most systems have one star, but some, such as Sirius, glow brightly with two.



Binary system
Star systems that have two stars, such as Sirius, are called binary systems.

Brightest stars
Alpha Centauri is one of the brightest stars visible from Earth, apart from the sun. Others are Sirius and Procyon.





Far-flung solar system
Our solar system is about 26,000 light-years from the center of our galaxy.

Galaxy
Galaxies are enormous groups of stars that are held together by the force of gravity but separated by distances millions of times bigger than the distances between planets in our solar system. Our sun is on one of the spiral arms of a galaxy called the Milky Way, which is 100,000 light-years across—small for a galaxy.

Units of measurement

Scientists measure quantities—such as length, mass, or time—using numbers, so that their sizes can be compared. For each kind of quantity, these measurements must be in units that mean the same thing wherever in the world the measurements are made.

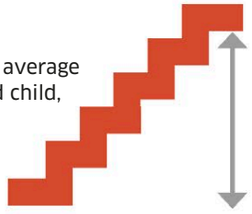
SI units
The abbreviation "SI" stands for *Système International*. It is a standard system of metric units that has been adopted by scientists all over the world so that all their measurements are done in the same way.

Base quantities

Just seven quantities give the most basic information about everything around us. Each is measured in SI units and uses a symbol as an abbreviation. The SI system is metric, meaning that smaller and larger units are obtained by dividing or multiplying by 10, 100, 1,000, etc. Centimeters, for instance, are 100 times smaller than a meter, but kilometers are 1,000 times bigger.

LENGTH
SI unit: meter (m)

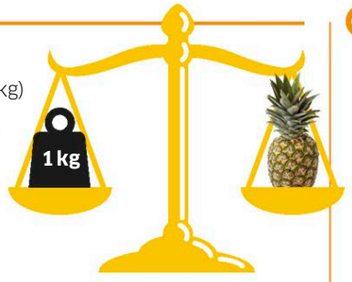
One meter is about the average height of a 3½-year-old child, or five steps up a typical staircase.



- A millionth of a meter (1 micrometer) = the length of a bacterium.
- A thousandth of a meter (1 millimeter) = the diameter of a pinhead.
- 1,000 meters (1 kilometer) = the average distance an adult walks in 12 minutes.

MASS
SI unit: kilogram (kg)

One kilogram is the mass of one liter of water, or about the mass of an average-sized pineapple.



- A thousand trillionth of a kilogram (1 picogram) = the mass of a bacterium.
- A thousandth of a kilogram (1 gram) = the mass of a paper clip.
- 1,000 kilograms (1 metric ton) = the average mass of an adult walrus.

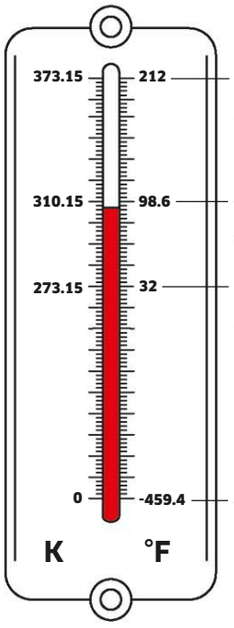
TIME
SI unit: second (s)

One second is the time it takes to swallow a mouthful of food, or to write a single-digit number.

- A thousandth of a second (1 millisecond) = the time taken by the brain to fire a nerve impulse.
- A tenth of a second (1 decisecond) = a blink of an eye.
- 1 billion seconds (1 gigasecond) = 32 years.

TEMPERATURE
SI unit: kelvin (K)

Just one degree rise in temperature can make you feel hot and feverish.



Temperature scales
In the USA, an everyday temperature scale uses degrees Fahrenheit (°F), where the freezing point of water is 32°F. Kelvin measures all the way down to absolute zero—where heat energy does not exist.

- 0 kelvin = absolute zero, when all objects and their particles are still.
- 1 kelvin = the coldest known object in the universe, the Boomerang Nebula.
- 1,000 kelvin = the temperature inside a charcoal fire.

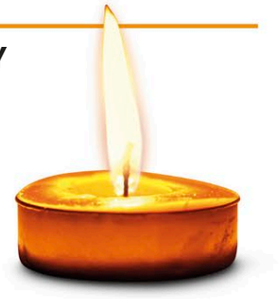
ELECTRICAL CURRENT
SI unit: ampere (A)

One ampere is about the current running through a 100 W light bulb.

- A thousandth of an ampere (1 milliampere) = the current in a portable hearing aid.
- 100,000 amperes = the current in the biggest lightning strikes.
- 10 thousand billion amperes = the current in the spiral arms of the Milky Way.

LIGHT INTENSITY
SI unit: candela (cd)

One candela is the light intensity given off by a candle flame.

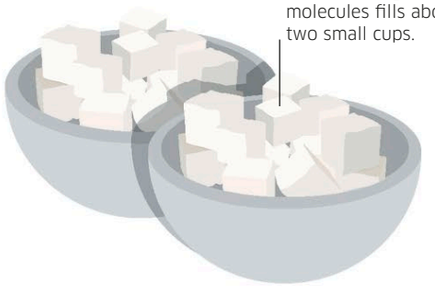


- A millionth of a candela = the lowest light intensity perceived by human vision.
- A thousandth of a candela = a typical night sky away from city lights.
- 1 billion candelas = the intensity of the sun when viewed from Earth.

AMOUNT OF A SUBSTANCE
SI unit: mole (mol)

One mole is a set number of atoms, molecules, or other particles. Because substances all have different atomic structures, one mole of one substance may be very different to that of another.

A mole of gold atoms is in about six gold coins.



A mole of sugar molecules fills about two small cups.

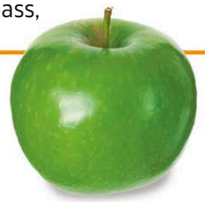
- A tenth of a mole of iron atoms = the amount of iron in the human body.
- 1,000 moles of carbon atoms = the amount of carbon in the human body.
- 10 million trillion moles of oxygen molecules = the amount of oxygen in Earth's atmosphere.

Derived quantities

Other kinds of quantities are also useful in science, but these are calculated from base quantities using scientific equations. For instance, we combine SI measurements of mass, distance, and time to work out an SI measurement for force. This means that force is said to be a derived quantity.

FORCE

SI unit: newton (N)



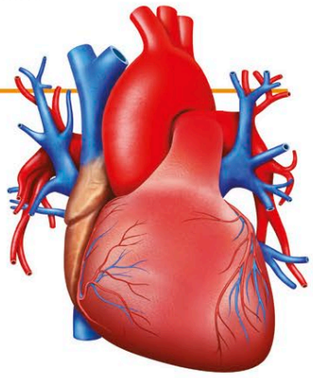
One newton is about the force of gravity on a single apple.

Force in newtons = **Mass in kilograms x distance in meters** / **Time in seconds^2**

- A 10 billionth of a newton = the force needed to break six chemical bonds in a molecule.
- 10 newtons = the weight of an object with mass of 1 kilogram.

FREQUENCY

SI unit: hertz (Hz)



One hertz is about the frequency of a human heartbeat: one beat per second.

Frequency in hertz = **Number of cycles** / **Time in seconds**

- 100 hertz = the frequency of an engine cycle in a car running at maximum speed.
- 10,000 hertz = the frequency of radio waves.

PRESSURE

SI unit: pascal (Pa)

One pascal is about the pressure of one bill of paper money resting on a flat surface.

Pressure in pascals = **Force in newtons** / **Area in meters ^{2}**

- A 10 thousand trillionth of a pascal = the lowest pressure recorded in outer space.
- 1 million pascals (1 megapascal) = the pressure of a human bite.

POWER

SI unit: watt (W)

One watt is about the power used by a single Christmas tree light.

Power in watts = **Energy in joules** / **Time in seconds**

- A millionth of a watt (1 microwatt) = the power used by a wristwatch.
- 1 billion watts (1 gigawatt) = the power used by a hydroelectric generating station.

ENERGY

SI unit: joule (J)

One joule is about the energy needed to lift a medium-sized tomato a height of one meter.

Energy in joules = **Force in newtons x distance in meters**

- A millionth of a joule (1 microjoule) = the energy of motion in six flying mosquitoes.
- 1,000 joules (1 kilojoule) = the maximum energy from the sun reaching 1 square meter of Earth's surface each second.

POTENTIAL DIFFERENCE

SI unit: volt (V)

Voltage is a measure of the difference in electrical energy between two points—the force needed to make electricity move. One volt is about the voltage in a lemon battery cell.

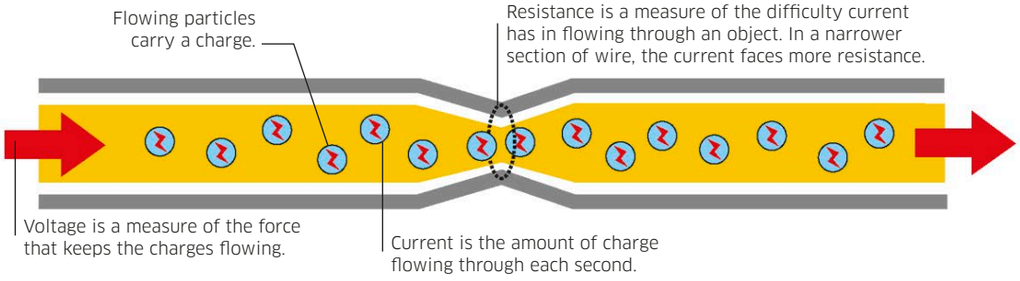
Potential difference in volts = **Power in watts** / **Current in amperes**

- 100 volts = the electrical grid voltage in the US.

ELECTRICAL CHARGE AND RESISTANCE

SI unit: Charge—coulomb (C)
Resistance—ohm (Ω)

The measurements relating to electricity are all interlinked. Charge is a measure of how positive or negative particles are, and can be calculated from the current and the time. Resistance is a measure of the difficulty a current has in flowing, and can be calculated from the voltage and the current.



Resistance in ohms = **Potential difference in volts** / **Current in amperes**

Charge in coulombs = **Current in amperes x time in seconds**

More than **half of all known animal species belong to the class of insects**, and most of these belong to one order: the beetles.

Classifying life

Scientists have described more than a million and a half different species of living things. They classify them into groups based on how they are related.

There are lots of ways of classifying organisms. Insects, birds, and bats could be grouped as flying animals, and plants could be grouped by how we use them. But neither of these systems shows natural relationships. Biological classification works by grouping related species. Bats, for instance, have closer links to monkeys than they do to birds, because they are both furry mammals that have evolved from the same mammal ancestors.

Single-celled organisms are the most common form of life in some kingdoms—including archaea, bacteria, and protozoans.

Seven kingdoms of organisms

- Archaea
- Bacteria
- Protozoans

More than 30 phyla of animals, including ...

- Flatworms
- Annelids
- Molluscs

12 classes of chordates, including ...

- Sea squirts
- Jawless fishes
- Cartilaginous fishes
- Lobe-finned fishes
- Ray-finned fishes

29 orders of mammals, including ...

- Monotremes
- Marsupials
- Elephants
- Sloths and anteaters
- Primates
- Rodents
- Rabbits, hares, and pikas

15 families of primates, including ...

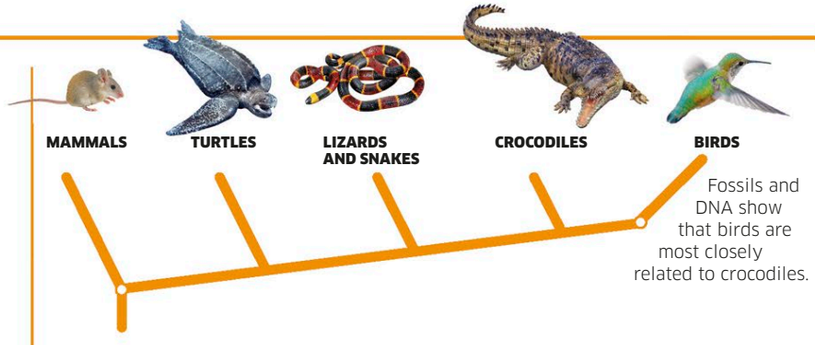
- Dwarf and mouse lemurs
- True lemurs
- Sifakas and relatives
- Bushbabies
- Aye-aye
- Lorises and relatives
- Tarsiers

Scientific names

Every species has a two-part scientific name using Latin words that are internationally recognized in science. The first part identifies its genus group, the second its species. Lions and tigers belong to the *Panthera* genus of big cats, but have different species names.

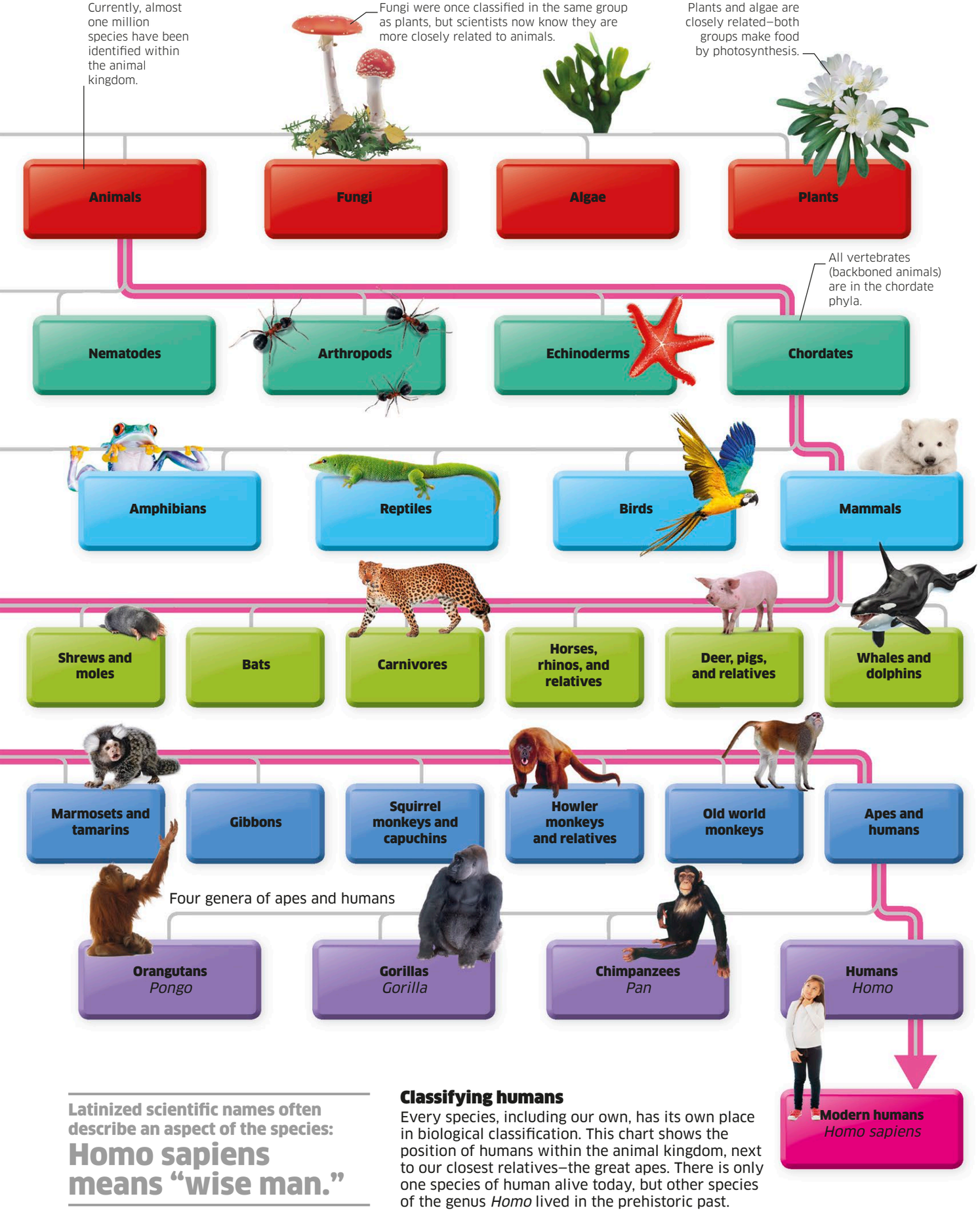
Panthera leo Lion

Panthera tigris Tiger



Classifying birds

Modern classification aims to show how organisms are linked by evolution. Birds and reptiles are traditionally in two separate classes. But birds evolved from reptile ancestors (see pp.136–137), so many scientists think they should be a sub-group of the reptiles.



Latinized scientific names often describe an aspect of the species:

Homo sapiens means "wise man."

Classifying humans

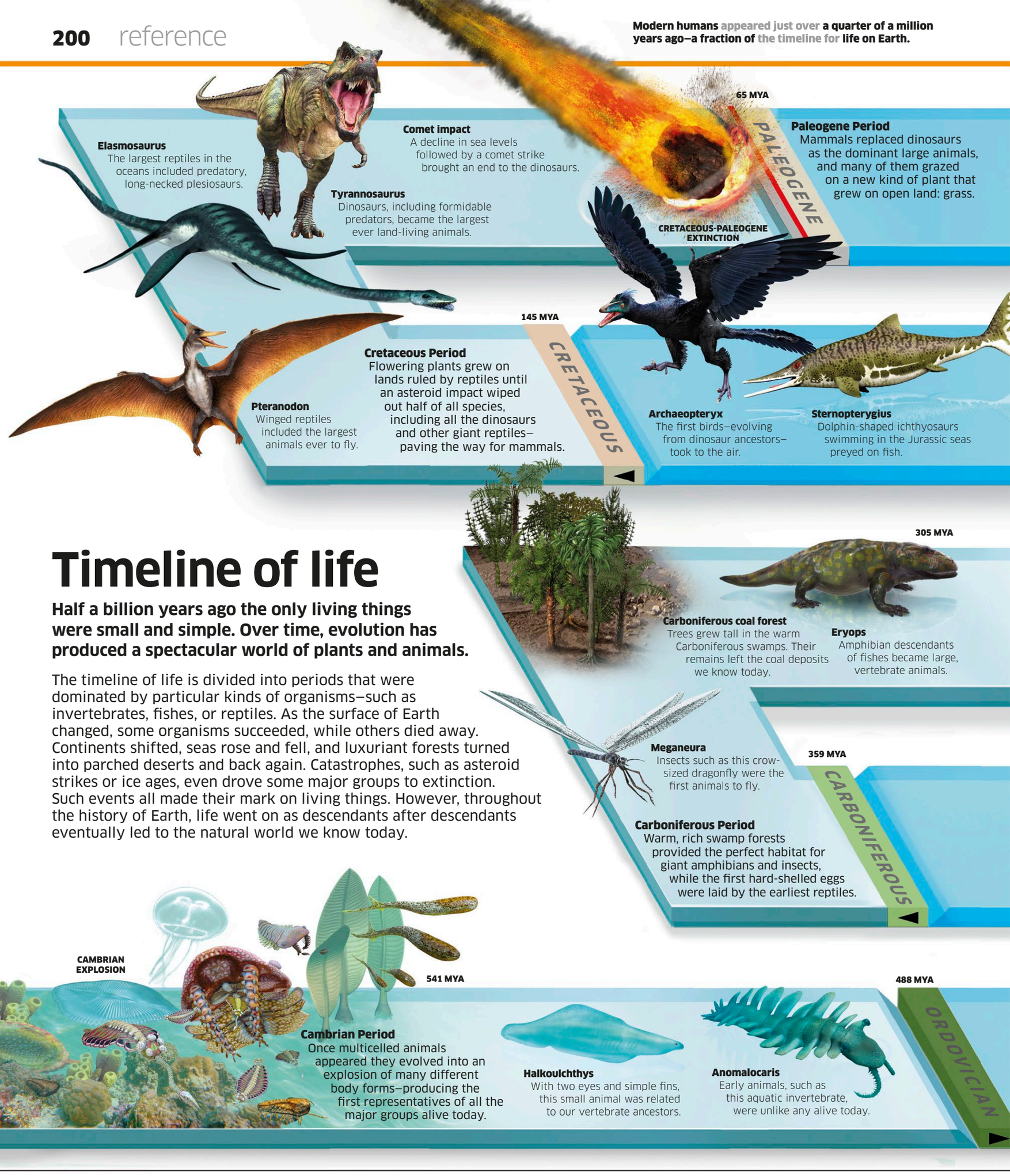
Every species, including our own, has its own place in biological classification. This chart shows the position of humans within the animal kingdom, next to our closest relatives—the great apes. There is only one species of human alive today, but other species of the genus *Homo* lived in the prehistoric past.

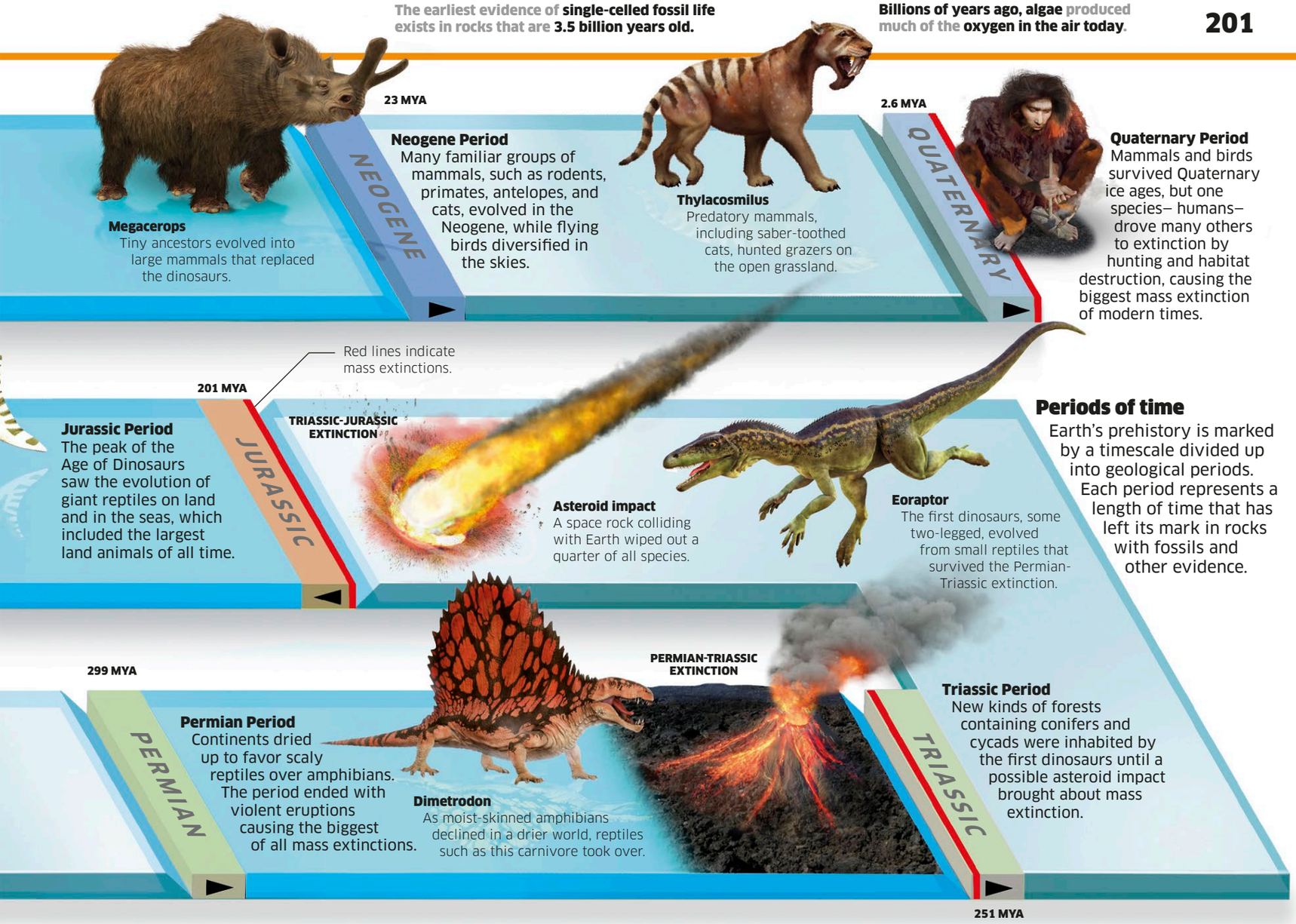
Modern humans appeared just over a quarter of a million years ago—a fraction of the timeline for life on Earth.

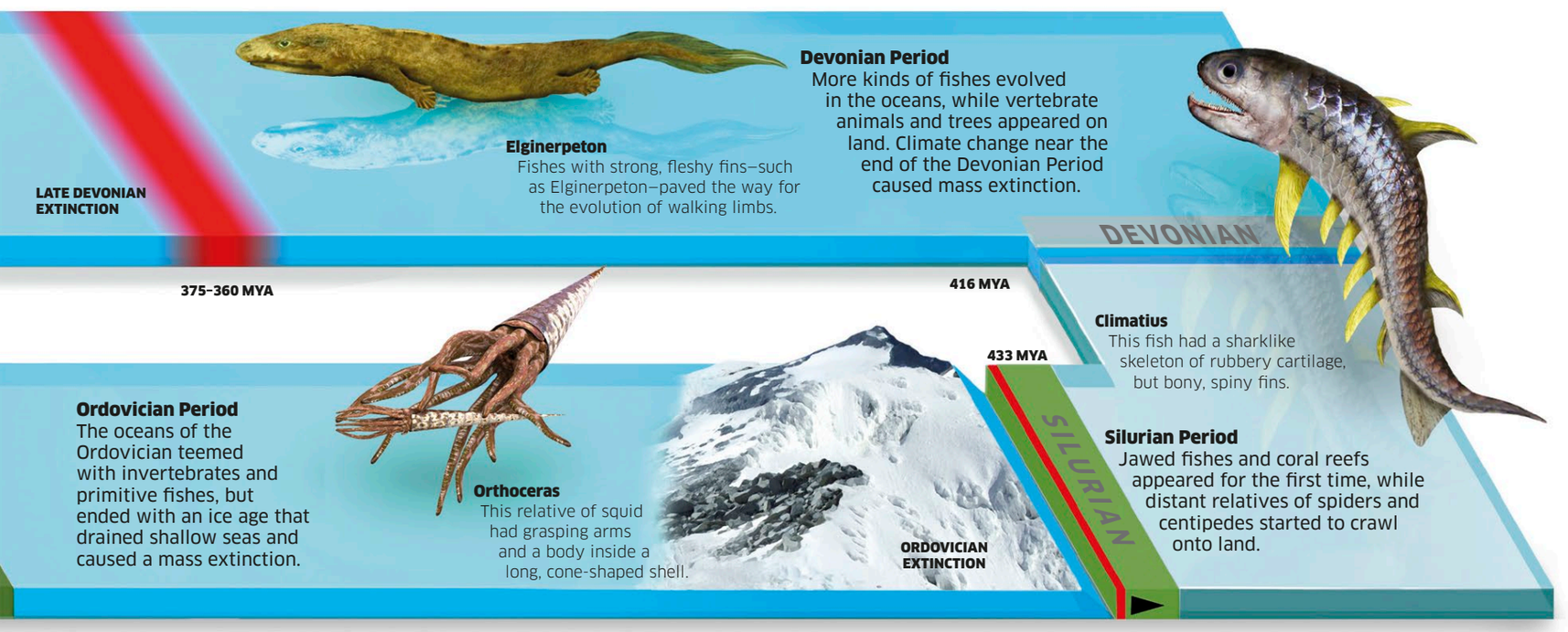
Timeline of life

Half a billion years ago the only living things were small and simple. Over time, evolution has produced a spectacular world of plants and animals.

The timeline of life is divided into periods that were dominated by particular kinds of organisms—such as invertebrates, fishes, or reptiles. As the surface of Earth changed, some organisms succeeded, while others died away. Continents shifted, seas rose and fell, and luxuriant forests turned into parched deserts and back again. Catastrophes, such as asteroid strikes or ice ages, even drove some major groups to extinction. Such events all made their mark on living things. However, throughout the history of Earth, life went on as descendants after descendants eventually led to the natural world we know today.




23 MYA
NEOGENE
Megacerops
Tiny ancestors evolved into large mammals that replaced the dinosaurs.
Neogene Period
Many familiar groups of mammals, such as rodents, primates, antelopes, and cats, evolved in the Neogene, while flying birds diversified in the skies.
Thylacosmilus
Predatory mammals, including saber-toothed cats, hunted grazers on the open grassland.
2.6 MYA
QUATERNARY
Quaternary Period
Mammals and birds survived Quaternary ice ages, but one species– humans– drove many others to extinction by hunting and habitat destruction, causing the biggest mass extinction of modern times.
Red lines indicate mass extinctions.
201 MYA
JURASSIC
Jurassic Period
The peak of the Age of Dinosaurs saw the evolution of giant reptiles on land and in the seas, which included the largest land animals of all time.
TRIASSIC-JURASSIC EXTINCTION
Asteroid impact
A space rock colliding with Earth wiped out a quarter of all species.
Eoraptor
The first dinosaurs, some two-legged, evolved from small reptiles that survived the Permian-Triassic extinction.
Periods of time
Earth's prehistory is marked by a timescale divided up into geological periods. Each period represents a length of time that has left its mark in rocks with fossils and other evidence.
299 MYA
PERMIAN
Permian Period
Continents dried up to favor scaly reptiles over amphibians. The period ended with violent eruptions causing the biggest of all mass extinctions.
Dimetrodon
As moist-skinned amphibians declined in a drier world, reptiles such as this carnivore took over.
PERMIAN-TRIASSIC EXTINCTION
TRIASSIC
Triassic Period
New kinds of forests containing conifers and cycads were inhabited by the first dinosaurs until a possible asteroid impact brought about mass extinction.
251 MYA


Elginerpeton
Fishes with strong, fleshy fins–such as Elginerpeton–paved the way for the evolution of walking limbs.
Devonian Period
More kinds of fishes evolved in the oceans, while vertebrate animals and trees appeared on land. Climate change near the end of the Devonian Period caused mass extinction.
LATE DEVONIAN EXTINCTION
DEVONIAN
375–360 MYA
416 MYA
Climatius
This fish had a sharklike skeleton of rubbery cartilage, but bony, spiny fins.
433 MYA
Ordovician Period
The oceans of the Ordovician teemed with invertebrates and primitive fishes, but ended with an ice age that drained shallow seas and caused a mass extinction.
Orthoceras
This relative of squid had grasping arms and a body inside a long, cone-shaped shell.
ORDOVICIAN EXTINCTION
SILURIAN
Silurian Period
Jawed fishes and coral reefs appeared for the first time, while distant relatives of spiders and centipedes started to crawl onto land.

Glossary

ACID
A substance with a pH lower than 7.

ALGAE
Plantlike organisms that can make food using energy from sunlight.

ALKALI
See Base

ALLOY
A mixture of two or more metals, or of a metal and a non-metal.

ANALOG
Relating to signals or information represented by a continuously varying value, such as a wave.

ATMOSPHERE
The layer of breathable gases, such as oxygen and nitrogen, that surrounds Earth.

ATOM
The smallest unit of an element.

BACTERIA
Microscopic organisms with a simple, single-celled form.

BASE
A substance with pH higher than 7. Bases that are soluble in water are called alkalis. Also: one of the four chemicals that make up the "rungs" of a DNA double helix.

BIOLOGY
The science of living things.

BUOYANCY
The tendency of a solid to float or sink in liquids.

CARBOHYDRATE
An energy-rich substance, such as sugar or starch.

CATALYST
A substance that makes a chemical reaction occur much more rapidly, but is not changed by the reaction.

CELL
The smallest unit of life.

CHEMICAL BOND
An attraction between particles, such as atoms or ions.

CHEMICAL REACTION
A process that changes substances into new substances by breaking and making chemical bonds.

CHEMISTRY
The science of matter and elements.

CHROMOSOME
A threadlike structure, found in the nucleus of cells, that is made up of coiled strands of DNA. Humans have 46 chromosomes per body cell.

CLIMATE
The most common weather conditions in an area over a long period of time.

COMBUSTION
A chemical reaction in which a substance reacts with oxygen, releasing heat and flames.

COMPOUND
A chemical substance in which two or more elements have bonded together.

CONCENTRATION
The amount of one substance mixed in a known volume of the other.

CONDENSATION
A process whereby a gas changes into a liquid.

CONDUCTOR
A substance through which heat or electric current flows easily.

COVALENT BOND
A type of chemical bond in a molecule where atoms share one or more electrons.

DNA
A material found in the cells of all organisms that carries instructions for how a living thing will look and function.

DRAG
The resistance force formed when an object pushes through a fluid, such as air or water.

ECOSYSTEM
A community of organisms and the nonliving environment around them.

ELECTRIC CHARGE
How positive or negative a particle is.

ELECTRON
One of the tiny particles inside an atom. It has a negative electric charge.

ELEMENT
A simple substance made of atoms that are all the same kind.

ENERGY
What enables work to be done. Energy exists in many different forms and cannot be created or destroyed, only transferred.

ENZYME
A substance produced in living organisms that acts as a catalyst and speeds up chemical reactions.

EROSION
A process by which Earth's surface rocks and soil are worn away by wind, water, or ice.

EVAPORATION
A process by which a liquid changes into a gas.

EVOLUTION
The process by which Earth's species gradually change over long periods of time, such as millions of years, to produce new species.

EXCRETION
The process by which living organisms expel or get rid of waste produced by cells of the body.

FERTILIZATION
The joining of male and female sex cells so they develop into new life.

FISSION
A splitting apart; nuclear fission is the splitting of the nucleus of an atom.

FOSSIL
The preserved remains or impressions of life from an earlier time.

FOSSIL FUEL
A substance formed from the remains of ancient organisms that burns easily to release energy.

FRICTION
The dragging force that occurs when one object moves over another.

FUSION
A joining together; nuclear fusion is the joining of two atomic nuclei.

GAS
A state of matter that flows to fill a container, and can be compressed.

GENE
One of the tiny units carried on DNA that determine what a living thing looks like and how it functions.

GLUCOSE
A simple carbohydrate, or sugar, made by photosynthesis and then used by cells as a source of energy.

GRAVITY
The force that attracts one object to another and prevents things on Earth from floating off into space.

HABITAT
The area where an animal naturally makes its home.

INHERITANCE
The range of natural characteristics passed on to offspring by parents.

INSULATOR
A material that stops heat moving from a warm object to a colder one.

ION
An atom that has lost or gained one or more electrons and as a result has either a positive or negative electric charge.

IONIC BOND
A type of chemical bond where one or more electrons are passed from one atom to another, creating two ions of opposite charge that attract each other.

ISOTOPE
One of two or more atoms of a chemical element that have different numbers of neutrons compared to other atoms of the element.

LIFT
The upward force produced by an aircraft's wings that keeps it airborne.

LIQUID
A state of matter that flows and takes the shape of a container, and cannot be compressed.

MAGMA
Hot, liquid rock that is found beneath Earth's surface.

MAGNET
An object that has a magnetic field and attracts or repels other magnetic objects.

MASS
A measure of the amount of matter in an object.

MATERIAL
A chemical substance out of which things can be made.

METAL
Any of many elements that are usually shiny solids and good conductors of electricity.

MICROORGANISM
A tiny organism which can only be seen with the aid of a microscope. Also known as a microbe.

MINERAL
A solid, nonliving material occurring naturally and made up of a particular kind of chemical compound.

MOLECULE
A particle formed by two or more atoms joined by covalent bonds.

MONOMER
A molecule that can be bonded to other similar molecules to form a polymer.

NERVE
A fiber that carries electrical messages (nerve impulses) from one part of the body to another.

NEUTRON
One of the tiny particles in an atom. It has no electric charge.

NUCLEUS
The control center inside the cells of most living organisms. It contains genetic material, in the form of DNA. Also: the central part of an atom, made of protons and neutrons.

NUTRIENT
A substance essential for life to exist and grow.

ORBIT
The path taken by an object, for example, a planet, that is circling around another.

ORGAN
A group of tissues that makes up a part of the body with a special function. Important organs include the heart, lungs, liver, and kidneys.

ORGANISM
A living thing.

PARTICLE
A tiny speck of matter.

PHOTOSYNTHESIS
The process by which green plants use the sun's energy to make carbohydrates from carbon dioxide and water.

PHYSICS
The science of matter, energy, forces, and motion.

PIGMENT
A chemical substance that colors an object.

POLLEN
Tiny grains produced by flowers, which contain the male cells needed to fertilize eggs.

POLYMER
A long, chainlike molecule made up of smaller molecules connected together.

PRESSURE
The amount of force that is applied to a surface per unit of area.

PRODUCT
A substance produced by a chemical reaction.

PROTEIN
A type of complex chemical found in all living things, used as enzymes and in muscles.

PROTON
One of the tiny particles inside an atom. It has a positive electric charge.

RADIATION
Waves of energy that travel through space. Radiation includes visible light, heat, X-rays, and radio waves. Nuclear radiation includes subatomic particles and fragments of atoms.

RADIOACTIVE
Describing a material that is unstable because the nuclei of its atoms split to release nuclear radiation.

REACTANT
A substance that is changed in a chemical reaction.

REACTIVE
A substance that is likely to become involved in a chemical reaction.

RESPIRATION
The process occurring in all living cells that releases energy from glucose to power life.

ROOM TEMPERATURE
A standard scientific term for comfortable conditions (for humans), usually a temperature of around 68°F (20°C).

SEX ORGANS
The organs of an organism that allow it to reproduce. They usually produce sex cells: sperm in males, and eggs in females.

SOLID
A state of matter in which an element's atoms are joined together in a rigid structure.

SOLUTE
A substance that becomes dissolved in another.

SOLVENT
A substance that can have other substances dissolved in it.

SYNTHETIC
Man-made chemical.

TISSUE
A group of similar cells that carry out the same function, such as muscle tissue, which can contract.

TOXIC
Causing harm, such as a poison.

ULTRASOUND
Sound with a frequency above that which the human ear can detect.

ULTRAVIOLET
A type of electromagnetic radiation with a wavelength shorter than visible light.

UNIVERSE
The whole of space and everything it contains.

VOLCANO
An opening in Earth's crust that provides an outlet for magma when it rises to the surface.

WAVE
Vibration that transfers energy from place to place, without transferring the matter that it is flowing through.

WAVELENGTH
The distance between wave crests, usually when referring to sound waves or electromagnetic waves.

WEIGHT
The force applied to a mass by gravity.